

Stochastic Gradient Descent

Smooth and strongly convex - need self tuning

$$\min: f(x) = \frac{1}{n} \sum f_i(x) \quad (\text{ERM})$$

Each f_i is β -smooth, f is α -strongly convex

Example: $\min_x \frac{1}{n} \|Ax - y\|_2^2 = \frac{1}{n} \sum_{i=1}^n \underbrace{(a_i^T x - y_i)^2}_{f_i}$

Reducing variance by recentering

$$f(x) = \frac{1}{n} \sum_{i=1}^n f_i(x) \quad : \quad \tilde{g}_x = \nabla f_I(x)$$

Observation : For any two points x, z again $I \sim \text{uniform } \{1, \dots, n\}$

$$\tilde{g}(x) = \nabla f_I(x) - (\nabla f_I(z) - \nabla f(z)) \quad \leftarrow \text{this is also a stoch. gradient}$$

$$\begin{aligned} \mathbb{E}_I[\tilde{g}(x)] &= \mathbb{E}[\nabla f_I(x)] - (\mathbb{E}[\nabla f_I(z)] - \nabla f(z)) \\ &= \nabla f(x) - (\nabla f(z) - \nabla f(z)) = \nabla f(x). \end{aligned}$$

The SVRG algorithm

Stochastic Variance-Reduced Gradient Descent -

Outer loop:

On the k^{th} iteration

$$x_1 = y_k$$

$$\left[\begin{array}{l} \text{Inner Loop: for } t = 1, \dots, T \\ x_{t+1} = x_t - \eta (\nabla f_I(x_t) - (\nabla f_I(y_k) - \nabla f(y_k))) \end{array} \right]$$

$$\text{Update: } y_{k+1} = \frac{1}{T} \sum_{t=1}^T x_t$$

Inner loop: each iteration requires same effort as previous SGD alg.
i.e., need to compute $\nabla f_I(x_t)$ — y_k does not change

Outer loop: Expensive: Requires full gradient computation,

Variance reduction lemma

Key lemma: Let $f_1, \dots, f_n : \beta$ -smooth, $I \sim \text{uniform}[n]$

Then:
$$\mathbb{E}_I \left[\|\nabla f_I(x) - \nabla f_I(x^*)\|_2^2 \right] \leq 2\beta (f(x) - f(x^*))$$

Note: $\nabla f_I(x)$ may not be small when $x = x^*$.

Proof of variance reduction lemma

Let: $g_i(x) = f_i(x) - [f_i(x^*) + \nabla f_i(x^*)^T (x - x^*)] \geq 0$ (by convexity)

Recall: h is β -smooth: $h(x - \frac{1}{\beta} \nabla h(x)) \leq h(x) - \frac{1}{2\beta} \|\nabla h(x)\|_2^2$

$$0 \leq g_i(x - \frac{1}{\beta} \nabla g_i(x)) \leq g_i(x) - \frac{1}{2\beta} \|\nabla g_i(x)\|_2^2$$

$$\Rightarrow -g_i(x) \leq -\frac{1}{2\beta} \|\nabla g_i(x)\|_2^2 \Rightarrow \underline{\underline{\|\nabla g_i(x)\|_2^2 \leq 2\beta g_i(x)}}$$

$$\|\nabla g_i(x)\|_2^2 = \|\nabla f_i(x) - \nabla f_i(x^*)\|_2^2 \leq 2\beta (f_i(x) - f_i(x^*) + \nabla f_i(x^*)^T (x - x^*))$$

$$\Rightarrow \boxed{\mathbb{E}_I [\|\nabla f_I(x) - \nabla f_I(x^*)\|_2^2]} \leq 2\beta (f(x) - f(x^*)) + \cancel{\nabla f(x^*)^T (x - x^*)} \rightarrow 0$$

$$\left(\mathbb{E}_I [f_I(x)] = \sum_{i=1}^n \frac{1}{n} f_i(x) = f(x) \right)$$

SVRG Theorem

Thm: $f = \frac{1}{n} \sum f_i(x)$

f_i is β -smooth

f is α -sc.

SVRG alg with step size $\eta = \frac{1}{10 \cdot \beta}$

Inner loop size: $T = 10 \cdot (\beta/\alpha)$

$\Rightarrow$ After $s+1$ iterations of outer loop

$$\mathbb{E} f(y^{s+1}) - f(x^*) \leq 0.9^s (f(y^{(1)}) - f(x^*))$$

key features: ① Linear convergence

② β/α does not appear in the conv. rate.

Proof of SVRG Result

Enough to show that:

$$\mathbb{E} f(y^{s+1}) - f(x^*) \leq 0.9 (f(y^s) - f(x^*))$$

Recall: $y^{s+1} = \frac{1}{T} \sum x_t$, x_t is produced in s^{th} inner loop.
($x_t = x_t^{(s)}$, $y = y^s$).

$$\begin{aligned} \|x_{t+1} - x^*\|_2^2 &= \|x_t - \eta (\nabla f_{i_t}(x_t) - (\nabla f_{i_t}(y) - \nabla f(y))) - x^*\|_2^2 \\ &= \|x_t - x^*\|_2^2 - 2\eta \underbrace{(\nabla f_{i_t}(x_t) - \nabla f_{i_t}(y) + \nabla f(y))^\top}_{V_t} (x_t - x^*) \\ &\quad + \eta^2 \|V_t\|_2^2, \\ &= \underbrace{\|x_t - x^*\|_2^2}_{(A)} - 2\eta \underbrace{V_t^\top}_{(B)} (x_t - x^*) + \underbrace{\eta^2 \|V_t\|_2^2}_{(C)} \\ &\quad \text{variance term} \end{aligned}$$

Proof of SVRG Result - cont -

$$\textcircled{C} \mathbb{E}_{i_t} [\|v_t\|_2^2] = \mathbb{E}_{i_t} \left[\|\nabla f_{i_t}(x_t) - \nabla f_{i_t}(y) + \nabla f_{i_t}(x^*)\|_2^2 \right]$$

$$(a+b)^2 \leq 2a^2 + 2b^2$$

$$\leq 2 \mathbb{E}_{i_t} [\|\nabla f_{i_t}(x_t) - \nabla f_{i_t}(x^*)\|_2^2] + 2 \mathbb{E}_{i_t} [\|\nabla f_{i_t}(y) - \nabla f_{i_t}(x^*)\|_2^2]$$

will drop this

$$\leq 2 \mathbb{E}_{i_t} [\|\nabla f_{i_t}(x_t) - \nabla f_{i_t}(x^*)\|_2^2] + 2 \mathbb{E}_{i_t} [\|\nabla f_{i_t}(y) - \nabla f_{i_t}(x^*)\|_2^2]$$

expectation is 0

$$\mathbb{E} [\|Y - \mathbb{E}[Y]\|^2] \leq \mathbb{E} [\|Y\|^2]$$

use variance reduction lemma twice

$$\leq 4\beta (f(x_t) - f(x^*) + f(y) - f(x^*))$$

Proof of SVRG Result - cont -

$$\begin{aligned}
 \textcircled{B} \quad & 2\eta v_t^T (x_t - x^*) \\
 \mathbb{E} [2\eta v_t^T (x_t - x^*)] &= 2\eta \cdot \mathbb{E} [v_t^T] (x_t - x^*) \\
 &= 2\eta \underbrace{\nabla f(x_t)^T (x_t - x^*)}_{\substack{\nabla f(x_t) \\ \text{!}}} \\
 &\geq 2\eta (f(x_t) - f(x^*)).
 \end{aligned}$$

Putting it all together:

$$\begin{aligned}
 \mathbb{E}_{i_t} [\|x_{t+1} - x^*\|^2] &\leq \textcircled{A} - \textcircled{B} + \textcircled{C} \quad \underbrace{\hspace{1cm}} \quad \underbrace{\hspace{1cm}} \\
 &\leq \|x_t - x^*\|^2 - 2\eta (f(x_t) - f(x^*)) + \eta^2 4\beta (f(x_t) - f(x^*) + f(y) - f(x^*)) \\
 &= \|x_t - x^*\|^2 - 2\eta (1 - 2\eta\beta) (f(x_t) - f(x^*)) + \eta^2 4\beta (f(y) - f(x^*))
 \end{aligned}$$

Proof of SVRG Result - cont -

Iterating this inequality:

$$\mathbb{E}_{i_t} [\|x_{t+1} - x^*\|^2] \leq \|x_1 - x^*\|^2 - 2\eta(1-2\eta\beta) \mathbb{E} \sum_{k=1}^t (f(x_k) - f(x^*)) \\ + \underbrace{\eta^2 4\beta \cdot t}_{\checkmark} (f(y) - f(x^*))$$

① x_1 is y

② by α -s.c. of f , $\|y - x^*\|^2 \leq \frac{2}{\alpha} (f(y) - f(x^*))$

$$\underbrace{2\eta(1-2\eta\beta)}_{\checkmark} \left(\mathbb{E} f\left(\underbrace{\frac{1}{T} \sum x_t}_{y^{s+1}}\right) - f(x^*) \right) \leq \left(\frac{2}{\alpha} + \eta^2 4\beta \cdot T \right) \cdot \frac{1}{T} (f(y) - f(x^*))$$

$\uparrow$
 $f(y)$

$$\mathbb{E} f(y^{s+1}) - f(x^*) \leq 0.9 (f(y^s) - f(x^*))$$

